Columbia Econ Math Camp 2025

Convexity

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1. Convexity in Sets

Convex Sets

Definition 1 (Convex Sets)

In any real vector space V, set $S \subset V$ is said to be **convex set** iff for any **weight** $\lambda \in [0,1]$ and $\boldsymbol{x}, \boldsymbol{y} \in S$, we have that $\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y} \in S$.

- It only makes sense to speak of convex set in vector spaces since basic operations $+, \cdot$ are well defined.
- The term $\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}$ is **convex combination** of $\boldsymbol{x}, \boldsymbol{y}$.
- More generally, for vector $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n \in V$, a convex combination $\boldsymbol{c}$ is a **weighted sum** of these vectors, i.e. $\boldsymbol{c} = \sum_{i=1}^{n} \lambda_i \boldsymbol{v}_i$ for some scalars $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{R}_+$ s.t. $\sum_{i=1}^{n} \lambda_i = 1$.
- Visually (if $V = \mathbb{R}^n$) a set S is convex if the straight line that connects the endpoints of any two vectors in S is fully contained in S.

Convex Sets

The next proposition provides another characterization of convexity.

Proposition 1

In vector space V, the set $S \subset V$ is convex iff any convex combination of finitely many vectors in S is also in S.

Further, any arbitary intersection of convex sets is also a convex set

Proposition 2

In vector space V, let $\{S_\alpha\}_{\alpha \in A}$ be a family of convex sets. Then $\bigcap_{\alpha \in A} S_\alpha$ is also convex.

Convex Sets

Now let's define the notion of the *smallest convex set that contains a set*

Definition 2 (Convex hull)

In vector space V, the **convex hull** of set $S \subset V$, or $Co(S)$ is the intersection of all convex sets in V containing S. That is, let $\mathcal{C}(S) := \{X \subset V : X \text{ is convex and } S \subset X\}$, then $Co(S) := \bigcap_{C \in \mathcal{C}(S)} C$.

- The proposition from previous silde guarantees that $Co(S)$ is convex.
- Moreover, if S is finite then finding its convex hull is simple:

Proposition 3

In vector space V, let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n\} \subset V$, then

$$Co(S) = \left\{ \sum_{i=1}^{n} \lambda_i \boldsymbol{v}_i : \lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{R}_+ \text{ and } \sum_{i=1}^{n} \lambda_i = 1 \right\}$$

Theorems**

Definition 3 (Hyperplane)

In $\mathbb{R}^n$, a **hyperplane** is a parametrized set $H(\boldsymbol{p}, c) = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{p} \cdot \boldsymbol{x} = c\}$ for $\boldsymbol{p} \in \mathbb{R}^n - \{\mathbf{0}\}$ and $c \in \mathbb{R}$.

- A hyperplane splits $\mathbb{R}^n$ into two: $H^-(\boldsymbol{p}, c) = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{p} \cdot \boldsymbol{x} \leq c\}$ and $H^+(\boldsymbol{p}, c) = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{p} \cdot \boldsymbol{x} \geq c\}$.
- The next theorem, which is key in 2nd Welfare Theorem, says that two disjoint convex sets can be *separated* by some hyperplane.

Theorem 1 (Minkowski's separating hyperplane)

Let $S_1, S_2 \subset \mathbb{R}^n$ be disjoint, convex and nonempty. Then there exists some $\boldsymbol{p} \in \mathbb{R}^n - \{\mathbf{0}\}$ and $c \in \mathbb{R}$ s.t. either $S_1 \subset H^-(\boldsymbol{p}, c)$ and $S_2 \subset H^+(\boldsymbol{p}, c)$, or $S_2 \subset H^-(\boldsymbol{p}, c)$ and $S_1 \subset H^+(\boldsymbol{p}, c)$

Theorems**

There is also a famous result that we will not prove but is useful for you:

Theorem 2 (Brouwer's fixed point)

Let X be a nonempty, compact and convex set in $\mathbb{R}^n$, $f : X \to X$ is continuous. Then f has a fixed point, i.e. there exists some $\boldsymbol{x}^* \in X$ s.t. $f(\boldsymbol{x}^*) = \boldsymbol{x}^*$

This theorem is used in Game Theory to prove the existence of Nash Equilibriums, and in proving the existence of Walrasian equilibrium in microeconomics.

2. Convexity in Functions

Convex and Concave Functions

Now let's define convexity in functions.

Definition 4 (Convex and concave functions)

Let S be convex set in vector space V, $f : S \to \mathbb{R}$. Then f is...

1. ... **convex** iff $f(\lambda \boldsymbol{x} + (1-\lambda)y) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in [0,1]$.
2. ... **concave** iff $f(\lambda \boldsymbol{x} + (1-\lambda)y) \geq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in [0,1]$.
3. ... **strictly convex** iff $f(\lambda \boldsymbol{x} + (1-\lambda)y) < \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in (0,1)$ with $x \neq y$.
4. ... **strictly concave** iff $f(\lambda \boldsymbol{x} + (1-\lambda)y) > \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y})$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in (0,1)$ with $x \neq y$.

Visually, if a straight line joining any two points of the graph of f is below (above) the function's graph, then f is concave (convex).

Convex and Concave Functions

The definition generalizes to convex combination of finitely many points:

Theorem 3 (Jensen's Inequality)

Let S be convex set in vector space V. Then $f : S \to \mathbb{R}$ is concave iff for any vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n \in S$ and scalars $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{R}_+$ that add up to 1, we have

$$f\left(\sum_{i=1}^{n} \lambda_i \boldsymbol{x}_i\right) \leq \sum_{i=1}^{n} \lambda_i f(\boldsymbol{x}_i)$$

For convex functions, Jensen's inequality hols if you change $\leq$ to $\geq$ due to next property.

Exercise 1

Show that f is concave iff $-f$ is convex.

Convex and Concave Functions

Definition 5 (Linear functions)

Let V, W be two vector spaces. We say $f : V \to W$ is a **linear function (or operator)** iff

1. $f(\boldsymbol{x}_1 + \boldsymbol{x}_2) = f(\boldsymbol{x}_1) + f(\boldsymbol{x}_2)$ for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in V$.
2. $f(\lambda \boldsymbol{x}) = \lambda f(\boldsymbol{x})$ for any $\boldsymbol{x} \in V, \lambda \in \mathbb{R}$

- Many famous operations are linear: derivative, integral, limit, summation, etc.
- When $V, W \subset \mathbb{R}$, a linear function is simply $f(\boldsymbol{x}) = \sum_{i=1}^{n} a_i x_i$ for some $a_1, a_2, \ldots, a_n \in \mathbb{R}$.

Proposition 4

Let $V, W \subset \mathbb{R}$ be convex sets. Then $f : V \to W$ is linear iff f is concave and convex.

Convex and Concave Functions

Let's formally define the graphs of a function

Definition 6 (Graph, epigraph and subgraph)

Let $f : S \to \mathbb{R}$, we define its **graph** as

$$G(f) := \{(x, y) \in S \times \mathbb{R} : y = f(x)\}$$

and its epigraph, $G^+(f) := \{(x, y) \in S \times \mathbb{R} : y \geq f(x)\}$, and subgraph, $G^-(f) := \{(x, y) \in S \times \mathbb{R} : y \leq f(x)\}$.

- Whereas the graph of f is our common notion of the function's plot, its epigraph is the entire region *above the graph*.
- The subgraph is the region *below the graph*.
- We can link properties of these two to convexity using the following proposition.

Convex and Concave Functions

Proposition 5

Let S be convex set in vector space V. We have $f : S \to \mathbb{R}$ is

1. ... convex iff $G^+(f)$ is a convex set in $V \times \mathbb{R}$.
2. ... concave iff $G^-(f)$ is a convex set in $V \times \mathbb{R}$.

Further, it is straightforward to notice the following

Exercise 2

Show that $f + g$ is concave function if f and g are each concave. Further, for $c \in \mathbb{R}$ show that cf is concave when f is concave.

Clearly these both would hold if we replaced *concave* with *convex*.

Convex and Concave Functions

Concavity and convexity are preserved under monotonic and convex transformations too.

Proposition 6

Let S be convex set in vector space V, $f : S \to \mathbb{R}$. We have

1. If f is convex and $\phi : \mathbb{R} \to \mathbb{R}$ is weakly increasing and convex, then $\phi \circ f$ is convex.
2. If f is concave and $\phi : \mathbb{R} \to \mathbb{R}$ is weakly increasing and concave, then $\phi \circ f$ is concave.

Exercise 3

Show that (i) a weakly decreasing concave transformation of a convex function is concave, and (ii) a weakly decreasing convex transformation of a concave function is convex.

Convex and Concave Functions

Definition 7 (Pointwise supremum and infimum)

Let $\mathcal{F}$ be a family of function from S to $\mathbb{R}$.

- If set $\{f(x) : f \in \mathcal{F}\} \subset \mathbb{R}$ is bounded from above for any $x \in S$, the **pointwise supremum** of $\mathcal{F}$ is $\sup \mathcal{F} : S \to \mathbb{R}$ with

$$\sup \mathcal{F}(x) := \sup\{f(x) : f \in \mathcal{F}\}$$

- If set $\{f(x) : f \in \mathcal{F}\} \subset \mathbb{R}$ is bounded from below for any $\boldsymbol{x} \in S$, the **pointwise infimum** of $\mathcal{F}$ is $\inf \mathcal{F} : S \to \mathbb{R}$ with

$$\inf \mathcal{F}(x) := \inf\{f(x) : f \in \mathcal{F}\}$$

Visually, a pointwise supremum is a function that any point x will output the maximum value $f(x)$ among all $f \in \mathcal{F}$. If the maximum is not well-defined, then it will output the supremum.

Convex and Concave Functions

Proposition 7

Let $\mathcal{F} = \{f_\alpha\}_{\alpha \in A}$ be a family of real valued functions on S which is a convex set in vector space V.

1. If all functions in $\mathcal{F}$ are convex, and set $\{f_\alpha(x) : \alpha \in A\} \subset \mathbb{R}$ is bounded from above for each $x \in S$, then the supremum function of $\mathcal{F}$ is convex.
2. If all functions in $\mathcal{F}$ are concave, and set $\{f_\alpha(x) : \alpha \in A\} \subset \mathbb{R}$ is bounded from below for each $x \in S$, then the supremum function of $\mathcal{F}$ is concave.

Crucially, this implies $\max\{f, g\}$ is convex when f, g are both convex, and $\min\{f, g\}$ is convex when f, g are both concave.

Convex and Concave Functions

Another characterization of convexity for C^1 functions.

Proposition 8

Let $f : S \to \mathbb{R}$ be C^1 , S is a convex and open set in $\mathbb{R}^n$. Then

(1) f is convex iff $f(\boldsymbol{x}') \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x}) \cdot (\boldsymbol{x}' - \boldsymbol{x})$, $\forall\, \boldsymbol{x}', \boldsymbol{x} \in S$.

(2) f is concave iff $f(\boldsymbol{x}') \leq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x}) \cdot (\boldsymbol{x}' - \boldsymbol{x})$ $\forall\, \boldsymbol{x}', \boldsymbol{x} \in S$.

(3) f is strictly convex iff $f(\boldsymbol{x}') > f(\boldsymbol{x}) + \nabla f(\boldsymbol{x}) \cdot (\boldsymbol{x}' - \boldsymbol{x})$ $\forall\, \boldsymbol{x}', \boldsymbol{x} \in S$ with $\boldsymbol{x}' \neq \boldsymbol{x}$.

(4) f is strictly concave iff $f(\boldsymbol{x}') < f(\boldsymbol{x}) + \nabla f(\boldsymbol{x}) \cdot (\boldsymbol{x}' - \boldsymbol{x})$ $\forall\, \boldsymbol{x}', \boldsymbol{x} \in S$ with $\boldsymbol{x}' \neq \boldsymbol{x}$.

Notice that the plot of RHS in all the inequalities is just a hyperplane that is tangent to the graph of f at $\boldsymbol{x}$. The proposition is telling us the plot of f must lie fully above/below this hyperplane iff f is convex/concave

Convex and Concave Functions

Proposition 9

Let $f : S \to \mathbb{R}$ be C^2 function with Hessian matrix $H(\boldsymbol{x})$ at any $\boldsymbol{x} \in S$, S is convex and open in $\mathbb{R}^n$.

(1) f is convex iff $H(\boldsymbol{x})$ is positive semi-definite for any $\boldsymbol{x} \in S$.

(2) f is concave iff $H(\boldsymbol{x})$ is negative semi-definite for any $\boldsymbol{x} \in S$.

(3) f is strictly convex iff $H(\boldsymbol{x})$ is positive definite for any $\boldsymbol{x} \in S$.

(4) f is strictly concave iff $H(\boldsymbol{x})$ is negative definite for any $\boldsymbol{x} \in S$.

We won't prove this formally, but a simple Taylor expansion says it all:

$$f(\boldsymbol{x}+\boldsymbol{h}) = f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})\boldsymbol{h} + \frac{1}{2}\boldsymbol{h}^T H_f(\boldsymbol{x})\boldsymbol{h} + o\left(\|\boldsymbol{h}\|^2\right)$$

When $\boldsymbol{h}$ is close to $\boldsymbol{0}$, $o\left(\|\boldsymbol{h}\|^2\right)$ is small compared to $\frac{1}{2}\boldsymbol{h}^T H_f(\boldsymbol{x})\boldsymbol{h}$, and therefore the sign of $f(\boldsymbol{x}+\boldsymbol{h}) - (f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})\boldsymbol{h})$ is determined by the sign of $\boldsymbol{h}^T H_f(x)\boldsymbol{h}$, which is a quadratic form!

3. Quasi-convexity and Quasi-concavity

Quasi-convexity and Quasi-concavity

Definition 8 (Quasi-convex and -concave functions))

Let S be a convex set in vector space V. $f : S \to \mathbb{R}$ is said to be:

(1) **quasi-convex** iff $f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \max\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in [0,1]$.

(2) **quasi-concave** if $f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \geq \min\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$ for any $\boldsymbol{x}, \boldsymbol{y} \in S, \lambda \in [0,1]$.

(3) **strictly quasi-convex** iff $f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) < \max\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$ for any $\boldsymbol{x}, \boldsymbol{y} \in S$ with $\boldsymbol{x} \neq \boldsymbol{y}$ and $\lambda \in (0,1)$.

(4) **strictly quasi-concave** iff $f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) > \min\{f(\boldsymbol{x}), f(\boldsymbol{y})\}$ for any $\boldsymbol{x}, \boldsymbol{y} \in S$ with $\boldsymbol{x} \neq \boldsymbol{y}$ and $\lambda \in (0,1)$.

- Concavity implies quasi-concavity, but the reverse is not true.
- Q-conc is actually sufficient for many results in optimization.

Quasi-convexity and Quasi-concave

The easiest way to visualize Q-conc is through contour sets.

Definition 9 (Contour Sets)

For $f : S \to \mathbb{R}$ we define the **upper contour set** of f with cutoff a as $C^+(f, a) := \{x \in S : f(x) \geq a\}$, and the **lower contour set** as $C^-(f, a) := \{x \in S : f(x) \leq a\}$

- From economics classes, you might remember the notion of a **level curve** of a function $f : S \to \mathbb{R}$. For a level $a \in \mathbb{R}$, which are all points x in domain S s.t. $f(x) = a$ (e.g. an indifference curve are all bundles that achieve a fixed utility level).
- For $a \in \mathbb{R}$, the upper contour set are all domain points that achieve a higher value in f than those in the level curve (e.g. bundles that achieve higher utility than the indifference curve).
- Contour sets are different from epigraphs and subgraphs!

Quasi-convexity and Quasi-concavity

The visual characterization of q-conc and q-conv is the following:

Proposition 10

Let S be a convex set in vector space V, $f : S \to \mathbb{R}$. Then

1. f is quasi-concave iff $C^+(f, a)$ is convex set in V for all $a \in \mathbb{R}$.
2. f is quasi-convex iff $C^-(f, a)$ is convex set in V for all $a \in \mathbb{R}$.

Finally, weakly-increasing transformations preserve q-conc and q-conv:

Proposition 11

Let S be a convex set in vector space V, $f : S \to \mathbb{R}$, $\phi : \mathbb{R} \to \mathbb{R}$ is weakly-increasing:

1. If f is quasi-concave then $\phi \circ f$ is quasi-concave.
2. If f is quasi-convex then $\phi \circ f$ is quasi-convex.